DSP Project Report

Dimensionality Reduction of University Dataset

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1 Problem Statement

The problem involves analyzing a university dataset containing multiple features related to aspects like applications, acceptance rates, and student information. After excluding the university type (e.g., Private University), the goal is to apply Principal Component Analysis (PCA) to reduce the 17 remaining features to a smaller set of components. The reduced data is visualized to capture key patterns and relationships in a simplified form.

2 Dataset Overview

2.1 Dataset Description

The dataset contains 19 features, described below:

- **Private:** Indicates if the university is private or public.
- **Apps:** Number of applications received.
- **Accept:** Number of applications accepted.
- **Enroll:** Number of new students enrolled.
- **Top10perc:** Percentage of new students from the top 10% of H.S. class.
- **Top25perc:** Percentage of new students from the top 25% of H.S. class.
- **F.Undergrad:** Number of full-time undergraduates.
- **P.Undergrad:** Number of part-time undergraduates.
- **Outstate:** Out-of-state tuition.
- **Room.Board:** Room and board costs.
- **Books:** Estimated book costs.
- **Personal:** Estimated personal spending.
- **PhD:** Percentage of faculty with PhDs.

- **Terminal:** Percentage of faculty with terminal degrees.
- **S.F.Ratio:** Student/faculty ratio.
- **perc.alumni:** Percentage of alumni who donate.
- **Expend:** Instructional expenditure per student.
- **Grad.Rate:** Graduation rate.

2.2 Visualization of Dataset

We used histograms to visualize parameter distributions and a heatmap to display correlations between features.

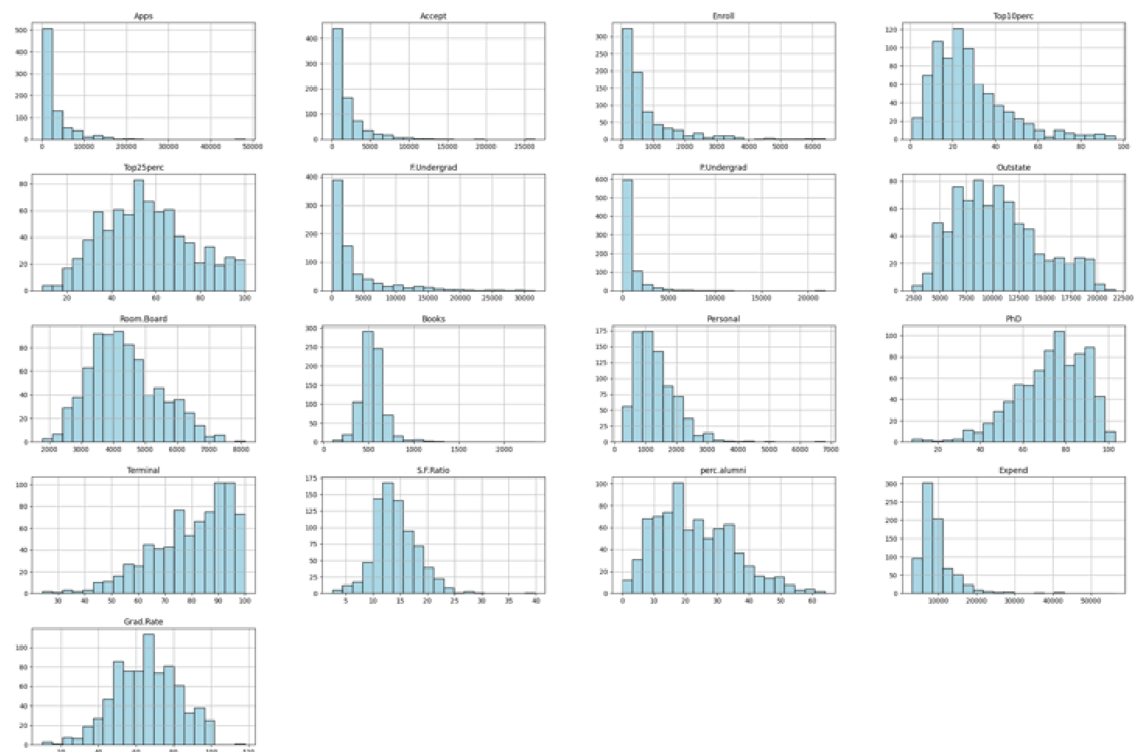


Fig 2.1 : Histogram of Parameters

Y Axis denotes frequency of Feature Values

X Axis represents the Value of each feature

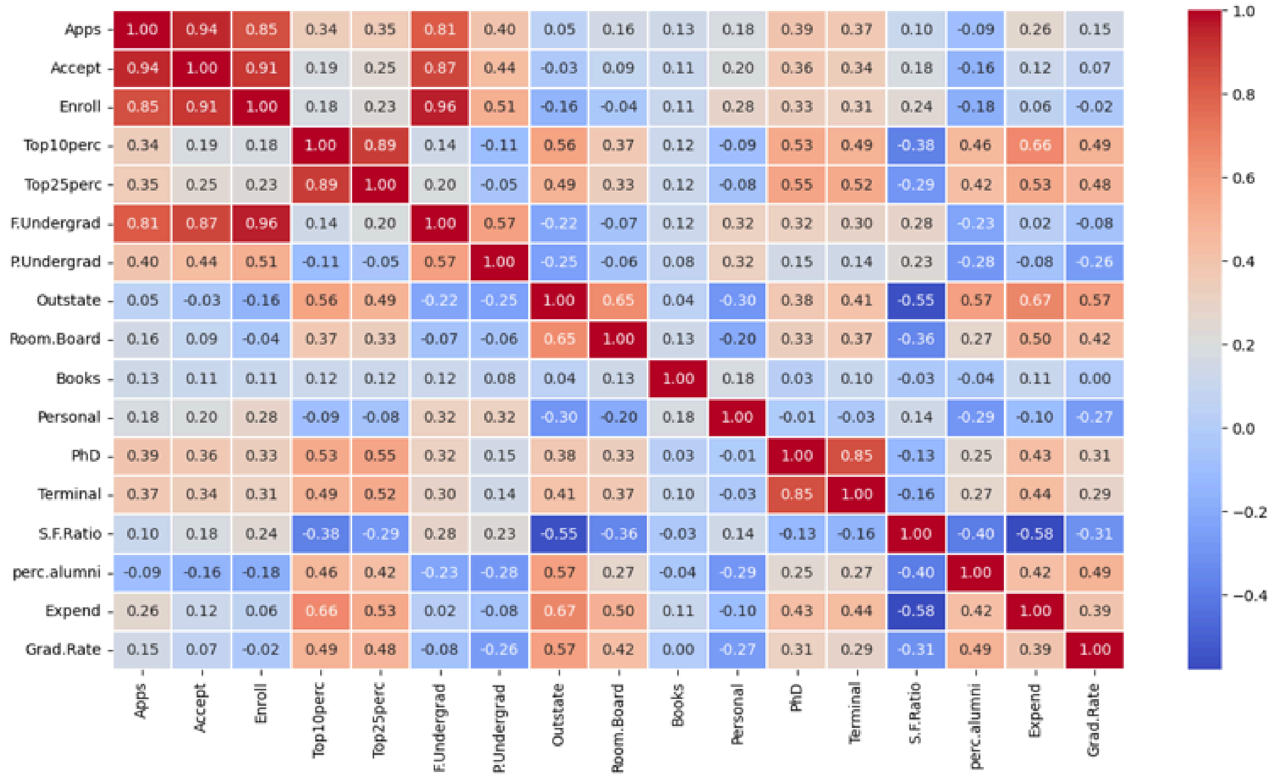


Fig 2.2: Correlation Matrix

3 Outlier Detection

3.1 Outlier 1: Graduation Rate

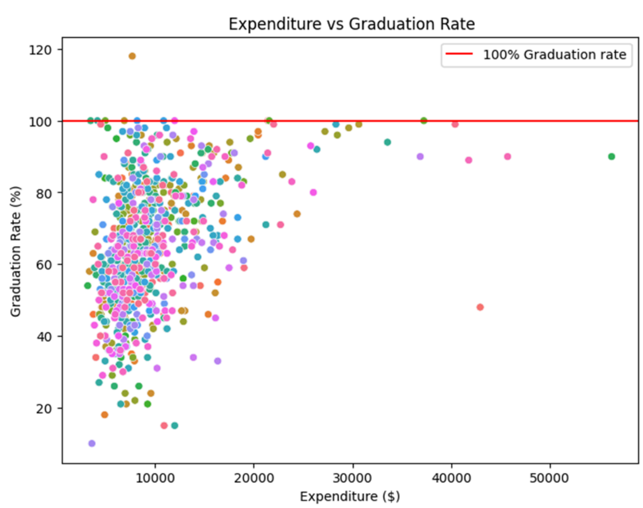


Fig 3.1 Graduation Rate Outlier

A university with a graduation rate greater than 100 was identified.

3.2 Outlier 2: Faculty with PhD Degrees

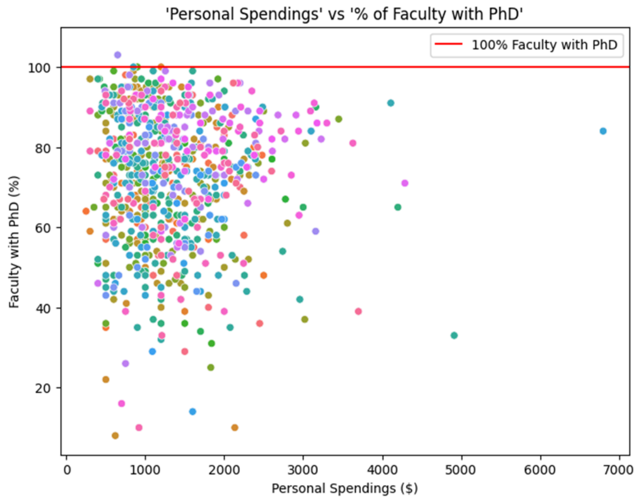


Fig 3.2 Faculty Phd Outlier

A university with over 100% of faculty having PhD degrees was identified.

4 Dimensionality Reduction

4.1 Principal Component Analysis (PCA)

PCA was used to reduce dimensionality while retaining the majority of data variance. The process involved:

1. Standardizing the data.
2. Calculating the covariance matrix.
3. Computing eigenvectors and eigenvalues.
4. Projecting the data onto principal components.

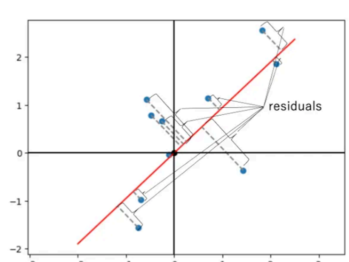


Fig 4.1 : example graph of PCA

4.2 Types of PCA applied:

4.2.1 Standard PCA

Transforms data into a new space by finding orthogonal components that maximize variance. Components are dense (use all features) and computationally efficient. Used for general dimensionality reduction and visualization.

4.2.2 Sparse PCA

Adds a sparsity constraint to PCA, producing components that use only a subset of features, improving interpretability. Useful for feature selection and analysis of high-dimensional data.

Aspect	Standard PCA	Sparse PCA
Goal	Maximize variance in the data	Maximize variance with sparsity in components
Mathematical Approach	Uses Singular Value Decomposition (SVD)	Adds L1 penalty for sparsity on coefficients
Interpretability	Components are linear combinations of all features	Components are sparse (few non-zero coefficients)
Efficiency	Can be computationally expensive with large datasets	Can be more compact and efficient with sparse components
Use Cases	Dimensionality reduction, noise filtering	When interpretability and feature selection are important
Computational Complexity	$O(n^2)$ for a data matrix of size $n \times p$	Higher than standard PCA due to sparsity constraints
Applications	General dimensionality reduction	Feature selection, interpretability (e.g., bioinformatics, text data)
Regularization	None	Includes L1 regularization to enforce sparsity

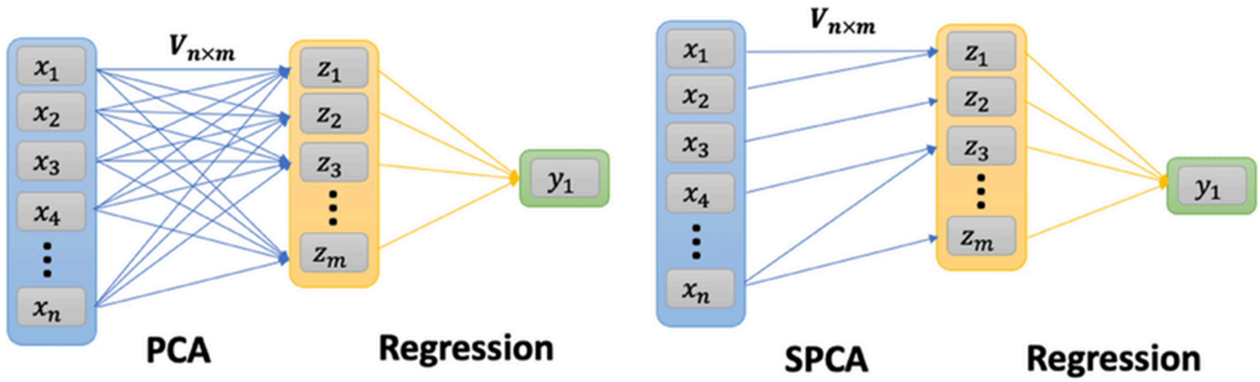
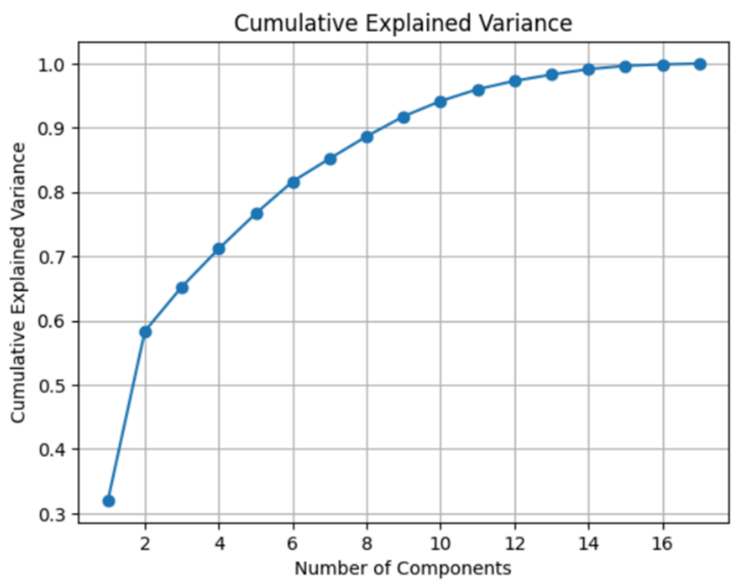


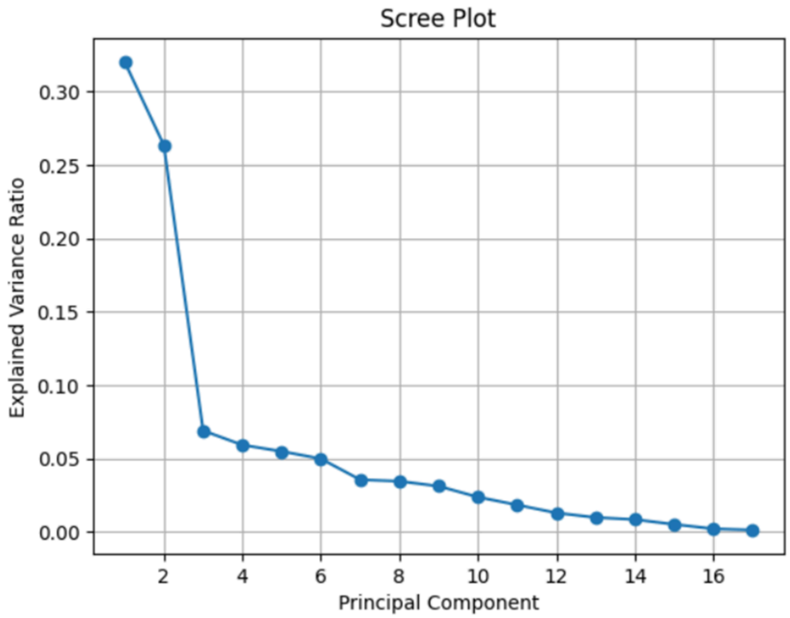
Fig: Sparse PCA

4.3 Results

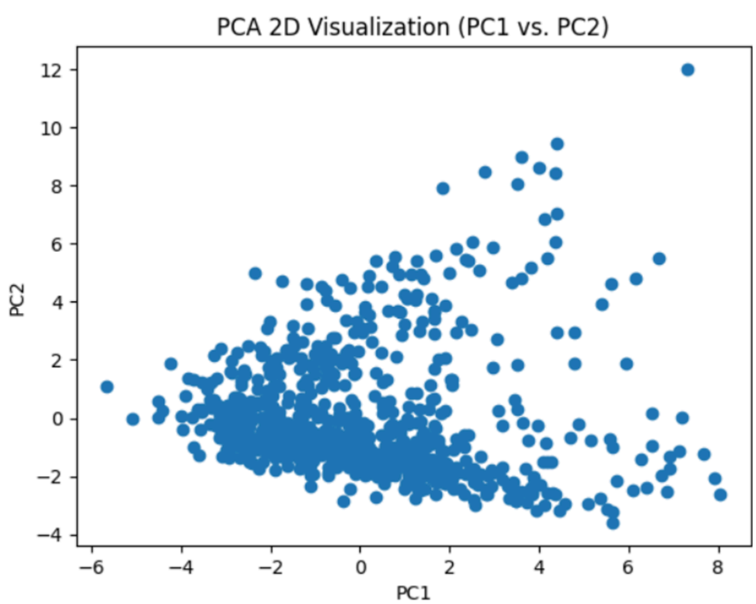
Component wise results for PCA :[Sheet](#) 🔗



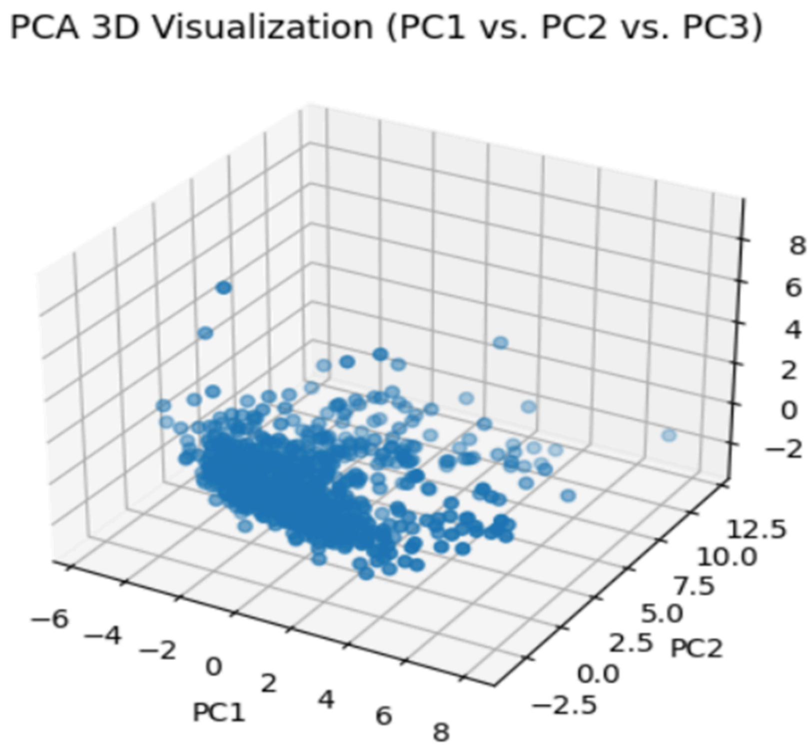
As the number of principal components taken increases,more variance is preserved.



The first principal component contributes maximum in explaining the variance of the original data. It corresponds to the highest eigenvalue of the covariance matrix. In general the order of contribution to explained variance is: PC1>PC2>PC3>.... and so on.



As we can see the points are close and there are few which are far away and they have outlier characteristics.



In the plot between PC1,PC2, PC3 also we can see that the points are close together and it indicates that the observations(rows in dataset) share similar relationships or characteristics based on the features that contribute most to principal components. So we have a pattern in the dataset.

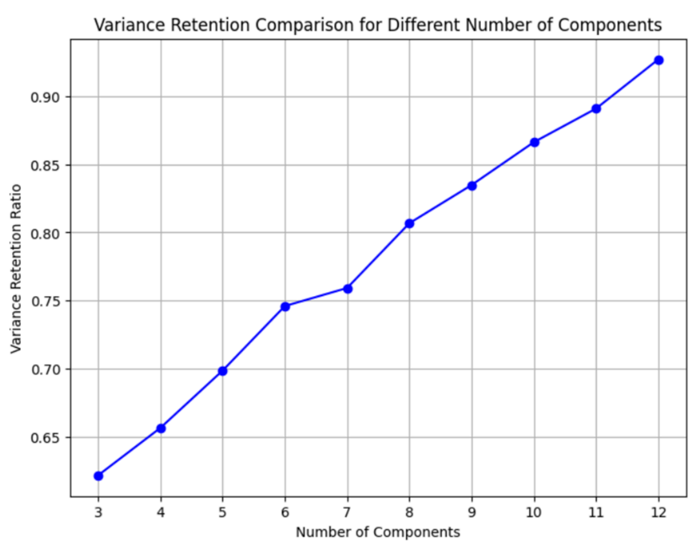


Fig: Sparse PCA Results

5 Challenges Faced

- Difficulty in visualizing high-dimensional data.
- Inability to use grid search for hyperparameter optimization due to the unsupervised nature of PCA.
- Small dataset size affecting correlation reliability.

6 Future Scope

- **Predictive Modeling:** Use reduced data to predict metrics like acceptance rates or graduation outcomes.
- **Integration with External Data:** Incorporate additional contextual information to enrich the analysis.

References

- Related Research Paper on Arxiv
- Mathematical Understanding of PCA on Medium

GitHUB Repository link